

run_pipeline.py — Vendor Intelligence Pipeline

Quality market-landscape pipeline. Works for ANY market — replace <Market> with your query.

START: `python run_pipeline.py --industry "<Market>" --country <global|Country> --live`

1 Phase 1 — Query Plan

LLM interprets the market query into a structured search plan: value-chain sections/segments to profile, geography scope, and search prompt seeds.

2 Phase 2 — Discovery + Entity Gate

Search-driven candidate discovery (web search, directories, AI Overview) across the value-chain sections. Drops geo-artifact / placeholder names (e.g. '<Country> <Market>') and obvious duplicates before enrichment.

3 Phase 3 — Enrichment

Crawls each candidate's website (smart_crawl / SSC) to extract facts: HQ location, founded year, revenue, YoY growth, product/service description, ownership.

4 Phase 4 — Classification (Quality)

LLM scores each company's relevance/confidence/quality for this market and tags a commercial_role (Brand, Marketer, Manufacturer, Solution Provider, etc.) from the enrichment evidence.

5 Export Filter + Save Landscape

Keeps rows clearing confidence/quality gates, scoped by config/industry_role_rules.yaml (per-market-family role allow-list) and the row cap (target: 1000 companies, global and regional runs). Writes CSV + Excel (.xlsx) + Word (.docx).

6 Coherent Quadrant Synthesis (synthesize_quadrant)

Takes the exported landscape rows and scores them on the Coherent Quadrant. Sub-steps below run in this order:

6a Select Industry Catalog — matches the market to an industry_group / industry_category baseline (config/quadrant_industry_criteria.yaml).

6b Define Market Axes — LLM refines the baseline into 5 X + 5 Y scoring parameters specific to THIS market. Axis TITLES are fixed for every market: X = 'Product Strength', Y = 'Business Strategy'. The 5 parameters under each axis change per market (axis_define.py).

6c Market Relevance / Player-Type Filter — LLM classifies the market as hardware / software_service / consumer, then keeps ONLY the matching role: hardware -> Manufacturer only; software_service -> Solution Developer / Service Provider / System Integrator only; consumer -> Brand / Marketer only (market_relevance.py).

run_pipeline.py — Vendor Intelligence Pipeline

Quality market-landscape pipeline. Works for ANY market — replace <Market> with your query.

6d Value-Chain Operator Filter — drops distributors/traders/resellers with no operational control, and collapses parent/subsidiary duplicates to one row (value_chain_filter.py).

6e Generate Scoring Questions — LLM writes 3 evidence-based questions per X/Y parameter (question_gen.py).

6f Score Each Company — builds a knowledge base from the enrichment evidence snapshot, then LLM answers the scoring questions per company (qa_scorer.py).

6g Normalize + Assign Quadrant — proportional score normalization across the population, then assigns Leaders / Challengers / Trailblazers / Emerging Players and a chart position (matrix_rollup.py, rating_map.py).

6h Build Chart + Table — chart shows ~20 brands split evenly across the four quadrants; Company Details table lists up to 1000 scored rows. Company column is never blank: shows the company's own name, plus '(acquired by Parent)' when owned (brand_meta.py).



7 Write Outputs

Quadrant JSON + Companies CSV + self-contained HTML report (output/quadrant/<market>_report.html) — chart, scorecards, Company Details table, and Market Scoring Parameters / Parameter Definitions panels.



END: CSV + XLSX + DOCX (landscape) + Quadrant JSON/CSV/HTML (chart)

Every LLM decision above (axis parameters, player-type, relevance, role, scoring) is re-run per market at call time — nothing here is hardcoded to one market.